ANALYSIS REPORT

Track 1 Remaining Family Partial Closeout Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the completed subset of the interrupted remaining-family exact-paper batch prepared in:

- `doc/reports/campaign_plans/track1/exact_paper/2026-04-18-00-48-05_track1_remaining_exact_paper_family_campaigns_plan_report.md`

  The remote terminal crashed during the `XGBM` portion of the sequence. The repository now has a clean partial-closeout state for the seven completed families and leaves `XGBM` plus `LGBM` explicitly pending for the follow-up crash diagnosis and rerun.
- completed family campaigns: `7`
- completed validation runs: `14`
- pending family campaigns: `2`
- completed families: `MLP`, `RF`, `DT`, `ET`, `ERT`, `GBM`, `HGBM`
- pending families: `XGBM`, `LGBM`
- aggregate campaign artifact root:

  `output/training_campaigns/track1_remaining_family_full_matrix_campaigns_2026_04_18_00_48_05/`

Operational Outcome

- all `14` finished runs produced validation summaries and Markdown validation reports
- the seven completed family campaign folders now expose `campaign_leaderboard.yaml`, `campaign_best_run.yaml`, and `campaign_best_run.md`
- the aggregate interrupted batch now also exposes the same three winner-bookkeeping artifacts under its canonical campaign root
- `XGBM` and `LGBM` are intentionally left open until the crash log is reviewed and the missing reruns are executed

Completed Family Closeout

Family	Amplitude Run	Phase Run	Best Run	Best Scope	Best Closure Score	Open Cells
DT	track1_dt_amplitude_full_matrix	track1_dt_phase_full_matrix	track1_dt_phase_full_matrix	phases_only	0.333	8
ERT	track1_ert_amplitude_full_matrix	track1_ert_phase_full_matrix	track1_ert_phase_full_matrix	phases_only	0.472	8
ET	track1_et_amplitude_full_matrix	track1_et_phase_full_matrix	track1_et_phase_full_matrix	phases_only	0.139	15
GBM	track1_gbm_amplitude_full_matrix	track1_gbm_phase_full_matrix	track1_gbm_phase_full_matrix	phases_only	0.556	5
HGBM	track1_hgbm_amplitude_full_matrix	track1_hgbm_phase_full_matrix	track1_hgbm_phase_full_matrix	phases_only	0.556	4
MLP	track1_mlp_amplitude_full_matrix	track1_mlp_phase_full_matrix	track1_mlp_amplitude_full_matrix	amplitudes_only	0.000	10
RF	track1_rf_amplitude_full_matrix	track1_rf_phase_full_matrix	track1_rf_phase_full_matrix	phases_only	0.611	3

Aggregate Ranking

Rank	Run	Family	Scope	Paper Cell	Met	Near	Open	Closure Score
1	track1_rf_phase_full_matrix	RF	phases_only	18	7	8	3	0.611
2	track1_gbm_phase_full_matrix	GBM	phases_only	18	7	6	5	0.556
3	track1_hgbm_phase_full_matrix	HGBM	phases_only	18	6	8	4	0.556
4	track1_ert_phase_full_matrix	ERT	phases_only	18	7	3	8	0.472
5	track1_hgbm_amplitude_full_matrix	HGBM	amplitudes_only	10	3	3	4	0.450
6	track1_ert_amplitude_full_matrix	ERT	amplitudes_only	10	2	3	5	0.350
7	track1_rf_amplitude_full_matrix	RF	amplitudes_only	10	1	5	4	0.350
8	track1_dt_phase_full_matrix	DT	phases_only	18	2	8	8	0.333
9	track1_gbm_amplitude_full_matrix	GBM	amplitudes_only	10	1	4	5	0.300
10	track1_et_phase_full_matrix	ET	phases_only	18	2	1	15	0.139
11	track1_dt_amplitude_full_matrix	DT	amplitudes_only	10	0	2	8	0.100
12	track1_et_amplitude_full_matrix	ET	amplitudes_only	10	0	2	8	0.100
13	track1_mlp_amplitude_full_matrix	MLP	amplitudes_only	10	0	0	10	0.000
14	track1_mlp_phase_full_matrix	MLP	phases_only	18	0	0	18	0.000

Partial Batch Winner

The aggregate bookkeeping winner is `track1_rf_phase_full_matrix` from family `RF`.

It wins the interrupted batch because it has the strongest closure score among the `14` completed runs while also keeping the fewest materially open cells under the explicit row-closure ranking policy.

- winning scope: `phases_only`
- closure score: `0.611`
- met paper cells: `7`
- near paper cells: `8`
- open paper cells: `3`

Scientific Outcome

- the seven rerun families materially refresh the row-reproduction evidence for `MLP`, `RF`, `DT`, `ET`, `ERT`, `GBM`, and `HGBM`
- several rows move numerically relative to the older `2026-04-14` full-matrix pass, so the canonical benchmark tables must now read from the refreshed family rows
- the partial refresh improves the envelope on a few exact-paper cells, but `Track 1` still remains open pending the unresolved harmonics and the missing `XGBM` / `LGBM` reruns
- `SVM` remains separately accepted as closed at the repository level with archived canonical reference models

Benchmark Refresh Effect

The canonical benchmark refresh after this partial closeout is:

- Table `2` amplitude `MAE`: `2/10` harmonics meet or beat the paper target
- Table `3` amplitude `RMSE`: `5/10` harmonics meet or beat the paper target
- Table `4` phase `MAE`: `5/9` harmonics meet or beat the paper target
- Table `5` phase `RMSE`: `5/9` harmonics meet or beat the paper target
- harmonic-level Table `6` closure: `2/10` fully matched, `5/10` partially matched, `3/10` not yet matched

  The highest-priority still-open harmonics after the refresh are now concentrated around:

- `3`
- `81`
- `156`
- `162`
- `240`

Pending Follow-Up

This partial closeout intentionally stops before any crash diagnosis. The next step is to inspect the terminal log, repair the interrupted remote sequence, and rerun the missing `XGBM` and `LGBM` family campaigns.